Keeping cool

The invention of refrigeration in the mid-19th century changed forever the way we eat and store food. The cold temperature in a refrigerator slows down the growth of bacteria that make food go bad, keeping items fresh for longer. An understanding of refrigeration led to the development of air conditioning, making it more comfortable to live in hot climates.

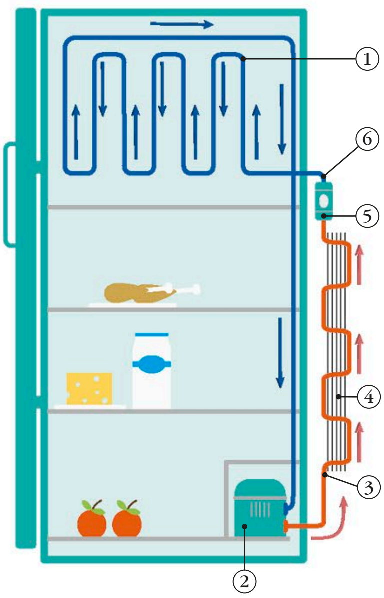
In this early fridge, the compressor unit that helps regulate the temperature is on the top. In most modern models, it is at the bottom.

A thick, insulated door keeps the inside cold.

The ice box is used for frozen food and ice cubes.

▸ **GE "MONITOR-TOP" REFRIGERATOR, 1934**
Invented by Christian Steenstrup of the American company General Electric, this was the world's first airtight refrigerator.

HOW A REFRIGERATOR WORKS



1. Coils absorb heat inside fridge.

2. Compressor squeezes the gas, heating it up as it leaves the fridge.

3. Gas travels through coils on the back of the fridge, cooling and turning back into a liquid.

4. Heat is radiated away from the fridge via the vent fins.

5. Expansion device expands the liquid coolant, turning it rapidly into a gas and making it cold.

6. The coolant goes back inside the fridge and the process repeats.

Refrigerators work by transferring heat from inside to outside the fridge. A substance called a coolant flows through the fridge in a set of coils. When the coolant is inside, it is cold and absorbs heat. As it leaves the refrigerator, it heats up and the heat is radiated away before the coolant re-enters the fridge.

CREATING A CHILL

Artificial refrigeration was invented by the Scottish physician William Cullen in the mid-18th century. It was not until 1899 that a patent was issued in the US to Arthur T. Marshall for the first mechanical refrigerator. The first fridge for home use, called the DOMELRE, was developed by the American engineer Fred W. Wolf, Jr. in 1913.